System Analysis & Design

Activity Diagram

Activity Diagram

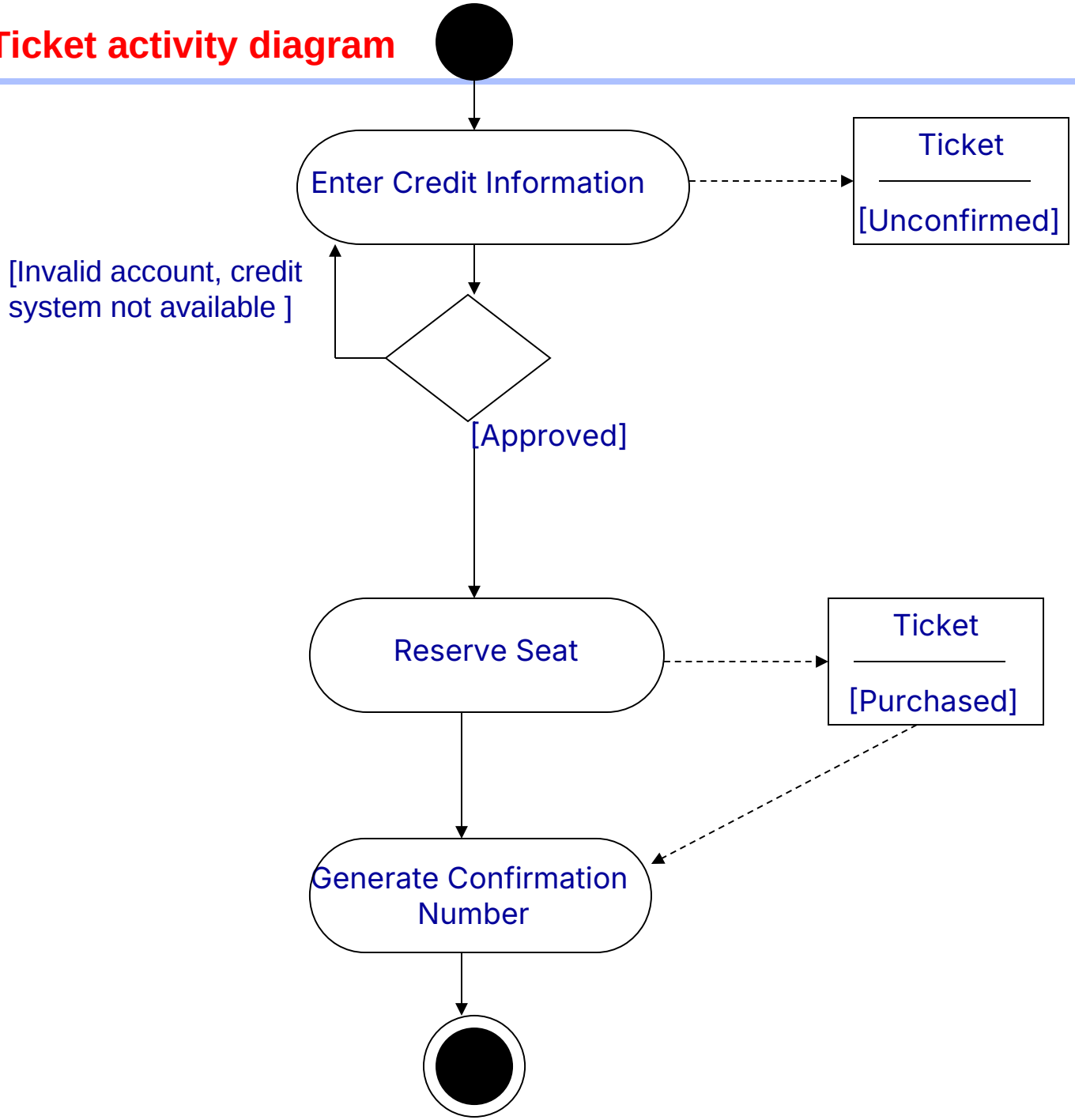
- Activity diagrams illustrate the *flow of functionality* in a system

- It used in requirements gathering to illustrate the *flow of events through a use case*

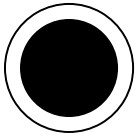
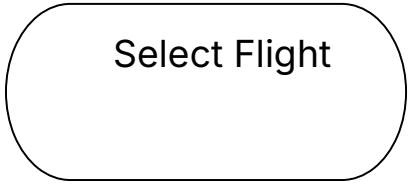
- These diagrams define *where the workflow starts*, *where it ends*, *what activities occur during the workflow*, and *in what order the activities occur*

- *An activity* is a task that is performed during the workflow

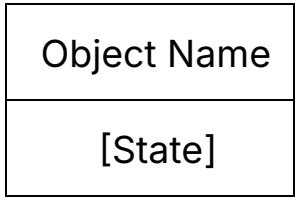
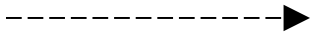
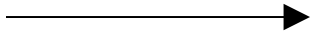
Purchase Ticket activity diagram



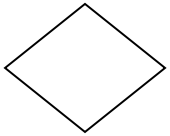
Elements of Activity Diagram

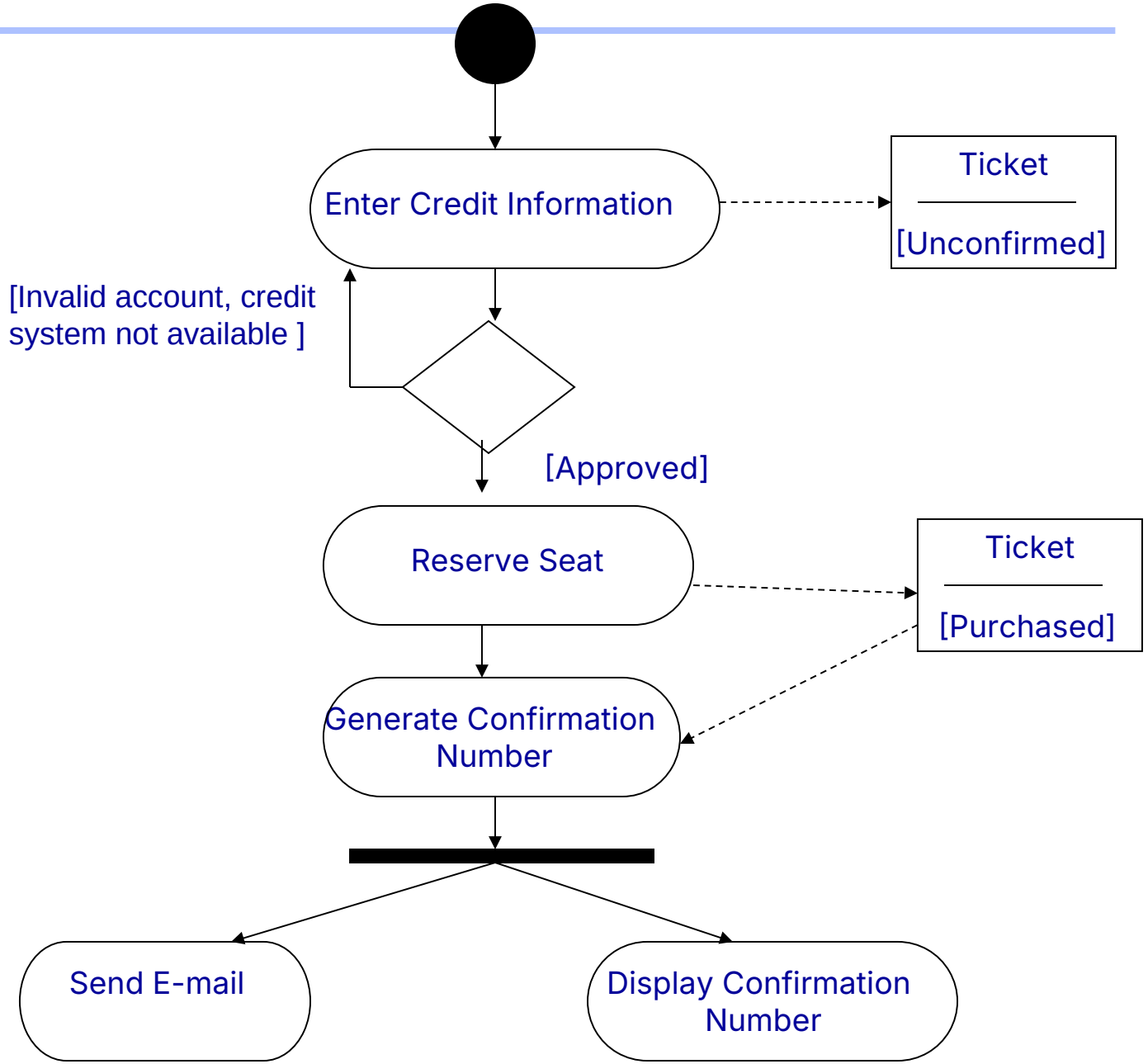
Element	Representation	Notation
Start State	- Show where the flow *begins* - Each activity diagram *must have one and only one* start state	
End State	- Show where the flow *ends*. - You can have more than one end state on the diagram (*Optional*)	
Activity	A step in the process - a task that is performed during the workflow	

Elements of Activity Diagram *cont*

Element	Representation	Notation
Objects	- An entity that activity is *used or changed* in the flow - You can understand where and how the object's state changes.	
Object Flows	- Link objects to activities *-From an activity to the object it changes* *-From the object to the activity that needs to use it (as input)*	
Transitions	shows how the *flow* of control moves from one activity to another	

Elements of Activity Diagram *cont*

Element	Representation	Notation
Synchronization	-A way to show that two or more branches of a flow occur in *parallel* - It can be either *horizontal* or *vertical*	
Swim lanes	- Each swimlane represents a different *role* within the workflow. - By looking at the activities within a given swimlane, you can find out the *responsibility of that role*	
Decision points	Shows *decision point* in the work flow	


Enter Credit Information
Ticket
[Unconfirmed]
[Invalid account, credit system not available]
[Approved]
Reserve Seat
Ticket
[Purchased]
Generate Confirmation Number
Send E-mail
Display Confirmation Number

Activity: *Actions*

- A detailed steps can be added to the activity by using *actions*. Actions are *smaller steps* that take place within an activity
- They may occur at one of four times:
  - *"entry" :* - Upon *entering* the activity
    - An entry action occurs as soon as the activity begins
  - *"exit":* - When *exiting* the activity
    - An exit action occurs as you are leaving the activity
  - *" do" :* - While performing the activity
    - These actions occur while in the activity and continue until you leave the activity
  - *" event"* : - followed by the event name
    - Upon a *specific* event. These actions happen *if and only if* a specific event occurs

Activity: *Actions* *cont*



Transitions: *Limitations*

- Event:
  - The event *must happen* in order for the transition to occur
  - The transition arrow is labeled with the event name, along with any arguments in parenthesis



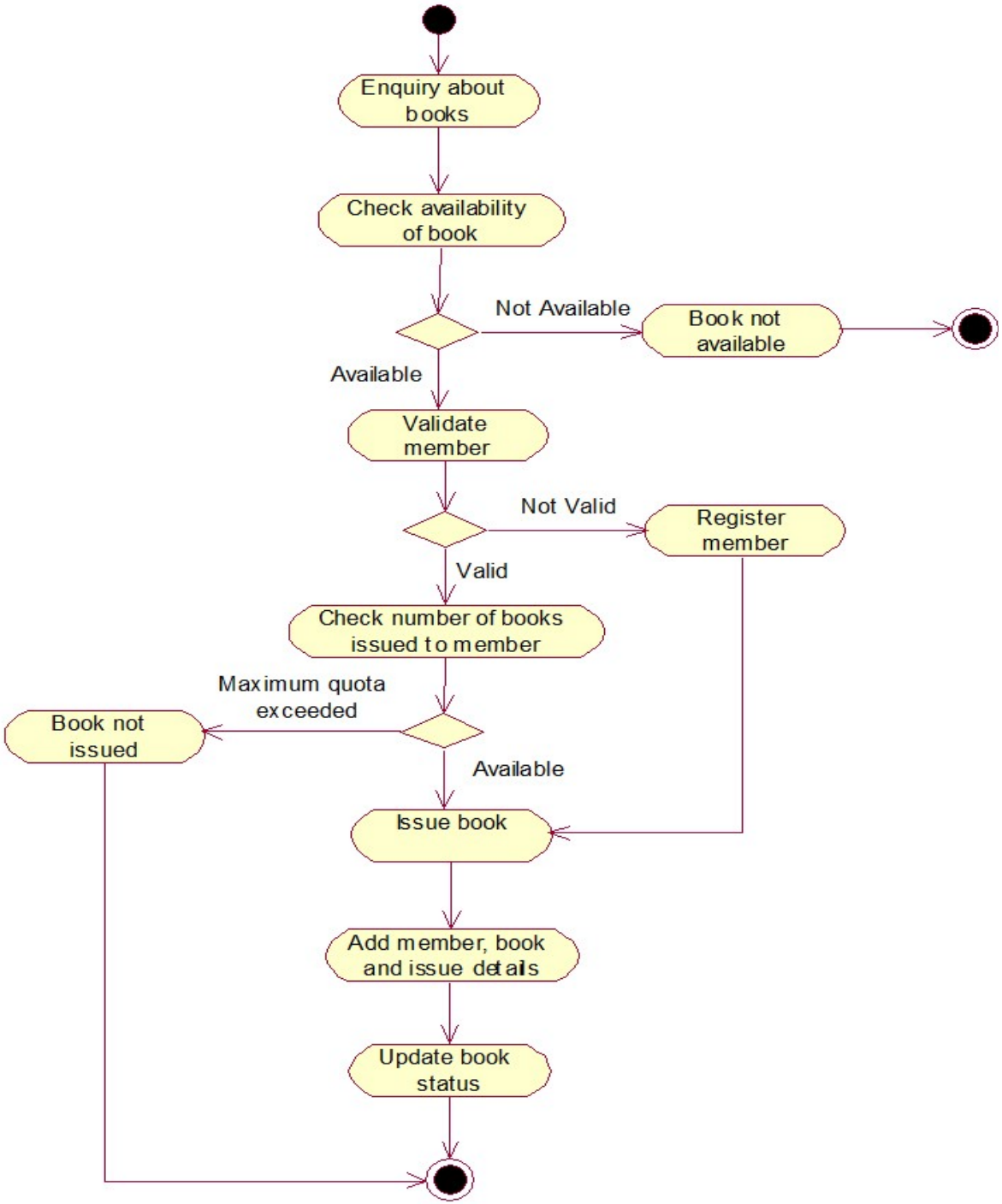
Transitions: *Limitations* *cont*

- Guard condition:
  - While an event triggers a transition, a guard condition controls *whether or not* the transition can occur
  - If a guard condition is present, it must be true in order for the transition to occur
  - The guard condition is listed along the transition arrow, following any event, and is enclosed in square brackets



Practice

- based on the *use case diagram* drawn last section for a *Library Management System*, Draw a detailed *activity diagram* for the *Borrow book/ Make Reservation use case*


Enquiry about books
Check availability of book
Not Available
Book not available
Available
Validate member
Not Valid
Register member
Valid
Check number of books issued to member
Maximum quota exceeded
Book not issued
Available
Issue book
Add member, book and issue details
Update book status

Thank You ☺